

Causal Inference for Spatial Treatments

Summary in brief

Pollmann, M. (in press). Causal inference for spatial treatments. *Econometrica*. Michael Pollmann, Department of Economics, Duke University.

The problem

Many events and policies happen at a specific location — a store opens, a plant is built, a local policy takes effect — and the question is their effect on nearby units. This is not the standard setting where each unit gets its own treatment, and the usual practice of comparing units by distance to the treated location does not answer it cleanly.

The reframing

- Start from the ideal experiment: what would we randomize if we could? Treatment *locations*, drawn from a set of plausible candidate sites.
- That implies comparing units near **realized** treatment locations with units near **counterfactual** candidate locations that went unused — not near-versus-far around the same site.
- This departs from current empirical practice, and it is the paper's central methodological point.

What the paper delivers

- **Design-based standard errors** that are straightforward to compute.
- **Inverse probability weighting estimators** comparing units near realized versus counterfactual locations.
- **Machine learning to find the counterfactual locations** in observational data — convolutional neural networks that learn the spatial distribution of covariates to identify observationally similar candidate sites.
- An extension of **double machine learning** to the design-based framework with spatial correlation, so high-dimensional covariates are admissible in the theory.

Key assumptions

- Outcomes are independent across candidate locations.
- Observable characteristics — not potential outcomes — determine treatment probabilities (an unconfoundedness-type condition).
- Treatment effects are additively separable across locations.

Empirical application

- Effect of grocery stores on foot traffic to nearby businesses during COVID-19 shelter-in-place policies, San Francisco Bay Area, using SafeGraph visitor data.
- Restaurants within roughly 0.025 miles of a real grocery store saw about double the visitors of those near counterfactual locations.
- The effect decays to zero beyond a few minutes' walk — large at very short distances, absent at larger ones.

Author's keywords

Causal inference; spatial treatments; design-based; ideal experiment; machine learning; convolutional neural networks.

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